Table 1. Demographic characteristics and main findings in the deaf population compared to normal hearing subjects.

References	Area of investigation	Method	Participants characteristics				Deafness characteristics				Statistical differences
			N°	Mean age ± SD (Range)	Sex (F/M)	Handedness	Severity (number)	Onset, language	Hearing aids	Etiology	
Allen J, 2008 *	insula	manual volumetry	25 D 16 HS 25 H	23,8 ± 4,1 (19-38) 24,3 ± 4,4 (19-38) 28,5 ± 4,5 (22-39)	38/28	RH RH RH	> 90 (21), > 75 dB (3), > 55 dB (1)	preL, SL	yes for some, after 2 years old	NS	GM: ↑ L post insula in D vs H WM: ↑ R insula in D and HS vs H
Allen J, 2013 *	CS, pars triangularis and opercularis, motor region	manual volumetry	25 D 16 HS 25 H	23,8 ± 4,1 (19-38) 24,3 ± 4,4 (19-38) 28,5 ± 4,5 (22-39)	38/28	RH RH RH	> 90 (21), > 75 dB (3), > 55 dB (1)	preL, SL	yes for some, after 2 years old	NS	GM: ↑ total CS in D vs H, ↑ L CS in D vs H, R CS larger than the L in all groups; ↑ bilat pars triangularis in D vs H and HS
Amaral L, 2016 *	Thal nuclei, IC, SC	manual volumetry	15 D 16 H	20,4 (17-22) 20,1 (18-22)	26/5	RH RH	> 90 dB	preL, SL	no	NS	GM: ↑ R vs L thal, lat gen nucl and IC
Emmorey K, 2003 *	temp lobe, STG, HG, PT	manual volumetry	25 D 25 H	23,8 ± 4,1 (19-38) 28,5 ± 4,5 (22-39)	28/22	RH RH	> 90 (21), > 75 dB (3), > 55 dB (1)	preL, SL	yes for some, after 2 years old	NS	GM: ↑ R STG and PT WM: ↓ bilat HG and STG
Feng G, 2018	WB	VBM and MVPS (GM, WM)	37 D 40 H	17,9 months ± 7,81 (8-38) 18 months ± 10,2 (8-38)	32/45	/ /	> 70 dB (27), > 50 dB (10)	preL, /	yes	NS	GM: ↓ bilat STG, HG, parahippocampal gyrus, L medial FG, L precuneus, L lingual gyrus, R supramarginal gyrus, R middle Cing, R SOG WM: ↓ bilat STG, front ant region, occ lobe, cerebellum
Fine I, 2005	early visual cortex (V1-V4), motion area (MT), defined by fMRI	VBM (GM, WM)	6 D 6 HS 6 H	27 ± 5,7 23,5 ± 6,3 26,8 ± 2,6	9/9	RH RH RH	> 80 dB	preL, SL	NS	NS	GM: no diff WM: no diff WM: no diff
Hribar M, 2014	HG, WB	CT, VBM, surface-based analysis (GM)	14 D 14 H	35,4 ± 6 (23-50) 30,5 ± 5,2 (23-50)	16/12	RH RH	> 90 dB	preL, SL	no	NS	GM: ↑ cerebellum WM: ↓ L HG
Kara A, 2006	CC	manual volumetry	18 D 18 H	41,2 ± 7,5 36,4 ± 7,1	0/36	RH RH	> 90 dB	preL, SL	NS	NS	GM: / WM: no diff
Kim D, 2009	WB	VBM (WM)	13 D 29 H	29,3 ± 6,8 26,5 ± 4,5	19/23	RH RH	> 90 dB	preL, NS	"not for brain developmental period"	fever, meningitidis, auditory nerve atrophy	GM: / WM: ↓ bilat STG and temp sub-gyral areas, L pariet, L SFG and L medial FG
Kim E, 2014	primary auditory cortex	VBM (GM)	8 D 11 D 11 H	50,4 ± 6,1 50,9 ± 12,2 49,5 ± 8,9	15/15	RH RH RH	> 70 dB >70 dB	preL, SL postL, oral	no yes	NS	GM: no diff
Kumar U, 2018	WB	VBM (GM, WM), source-based	50 D 50 H	19,5 19,5	matched but NS	matched but NS	> 90 dB	preL, oral	no	NS	GM: ↑ bilat MFG, bilat ITG, bilat fus, R cerebellum, L suppl motor area, R inf pariet region, ↑ CT bilat STG WM: ↓ bilat STG
Leporé N, 2010	CC, lobes, WB	tensor-based morphometry (GM, WM)	14 D 16 H	29,5 (21-52) 24	15/15	RH RH	> 90 dB	preL, SL	NS	NS but no CMV or meningitidis	GM: ↑ STG, R cerebellum, R MFG, L IFG, bilat preC WM: ↑ bilat HG and intraparietal areas
Li J, 2012 °	WB	VBM (GM, WM), CT	16 D 16 H	14,56 ± 2,10 (10-18) 14,75 ± 2,38 (10-18)	16/16	RH RH	> 90 dB	preL, SL	yes	drug toxicity (9), genetics (7)	GM: ↓ CT WB, L preC, R postC, L SOG and L fus gyrus WM: ↓ L middle front gyrus and R inf occ gyrus
Li W, 2013 °	WB	VBM (GM, WM), CT	16 D 16 H	14,56 ± 2,10 (10-18) 14,75 ± 2,38 (10-18)	16/16	RH RH	> 90 dB	preL, SL	yes	drug toxicity (9), genetics (7)	GM: ↑ Rw as of density in cerebellum, ↑ Lw as of CT in post Cing, ↑ Lw as of CT in gyri rectus (↑ L and ↓ R), ↑ Rw as of CT in precuneus, ↑ Rw as of CT of MFG (↓ L), ↑ Rw as of CT SFG (↓ L and ↑ R), ↑ Rw as of CT MOG WM: no diff
Li W, 2015 °	14 ROI within temp, front, parietal, occ gyri	manual volumetry	16 D 16 H	14,56 ± 2,10 (10-18) 14,75 ± 2,38 (10-18)	16/16	RH RH	> 90 dB	preL, SL	yes	drug toxicity (9), genetics (7)	GM: no diff
Meyer M, 2007	post Sylvian fissure, WB	manual curvature,	6 D	23,5 (19-43)	4/8	RH	NS	preL, SL	NS	NS	*sleeper slope of the post Sylvian fissure*

		VBM (GM, WM)	6 H	26,67 (22-45)		RH					WM: no diff
Olulade O, 2014 *	WB	VBM (GM, WM)	15 D	23,4 ± 3,3 (18,4-31,8)	32/28	RH	> 75 dB	preL, SL	NS	NS	GM: *D SL user vs HS* : ↑ R SFG, ↓ bilat STG, HG and insula, ↓R claustrum, R fus, R inf pariet and L MTG, D *vs H* : ↑ bilat MTG, SFG and R MFG, ↓ L ant fus, lingual gyri, cerebellum, and R post Cing, *SL vs English* : ↑ R medial FG, MFG and IFG and precuneus, L MFG and Cing
			15 D	28,2 ± 3,8 (22,8 - 34,6)		RH	> 75 dB	preL, oral			
			15 HS	26,7 ± 6,9 (18,4 - 39,5)		RH					WM: *D SL user vs HS* : ↓ bilat STG, HG, MTG, ITG, bilat fus and insula, ↓ L parahippocampal gyri; *D vs H* : ↓ L STG and HG; *SL vs English* : ↑ L preC and R IFG
			15 H	25,9 ± 6,0 (18,6 - 41,8)		RH					
Penhune V, 2003	HG, PT, WB	manual volumetry, VBM (GM, WM)	15 D	29	10/12	RH	> 90 dB	preL, SL	NS	NS	GM: no diff
			10 H	32		RH					WM: no diff
Pénicaud S, 2013	WB	VBM (GM, WM)	15 D	39,2 ± 12,3 (25-61)	33/33	RH	> 90 dB (18), > 80 dB (5)	preL, SL	NS	NS	GM: ↓ L V1, V2 and V3a/V7 in late SL acquisition, ↑ L V1, V2, V3a/V7 in SL acquisition during infancy
			43 H	37,3 ± 11,5 (25-62)		RH					WM: ↑ LV3a/V7 in late SL acquisition
Qi R, 2019	WB	VBM (GM)	35 D	39,72 ± 1,81	23/35	NS	> 90 dB (33), > 40 dB (2)	preL, NS	NS	Congenital (31), drug toxicity (2), infectious disease (2)	GM: ↓ R fus and R MOG
			23 H	39,83 ± 1,96		NS					WM: /
Shi B, 2016	WB	VBM (GM, WM)	15 D	9,36 ± 3,14 (5-13)	9/14	NS	> 90 dB	preL, SL	NS	NS	GM no diff
			12 H	8,41 ± 2,67 (5-14)		NS					WM: /
Shibata D, 2007	WB	VBM (GM, WM)	15 D	21 (18-27)	39/65	RH	> 90 dB	preL, SL	NS	most genetics and meningitidis	GM: no diff
			51 H	25 (19-32)		RH					WM: ↓ L HG
Shiohama T, 2019	WB	region and surface-based analysis (gyrification index, CT, surface, volume)	15 D	6,7 ± 5,2	16/32	NS	> 56 dB	preL, NS	yes (conventional or CI)	NS	GM: ↓ CT, surface areas and cortical volume of L MOG and L IOG
			36 H	6,7 ± 5		NS					WM: no diff
Smith K, 2011	HG, WB	manual volumetry,	15 D	14 months ± 3	26/16	/	> 70 dB (15), > 50 dB (1)	preL, /	NS	NS	GM: ↑ ant HG
			26 H	12 months ± 2,6		/					WM: ↓ ant HG, bilat STG, bilat cerebellum, R MTG, L temp, L post parietal, L occ
Smittenaar C, 2016	V1 (primary visula cortex), defined by fMRI	CT	15 D	39 ± 10,2	18/12	RH	severe to profound	preL, SL	NS	5 maternal rubella, 3 genetics, 6 unknown	GM: ↓ CT V1
			15 HS	38,32 ± 7,9		RH					WM: /

Note: * and °, same population in different papers; /, not investigated; WB, Whole Brain; VBM, Voxel Based Morphometry; GM, Grey Matter; WM, White Matter; CT, Cortical Thickness; D, Deaf; H, Hearing not signer; HS, Hearing Signers; RH, Right Handed; preL, PreLingual; PostL, PostLingual; NS, Not Specified; Thal, Thalamus; IC, Inferior Colliculi; SC, Superior Colliculi; CC, Corpus Callosum; D, Deaf; H, Hearing; SL, Sign Language; STG, Superior Temporal Gyrus; HG, Heschl's gyrus; PT, Planum Temporale; MTG, Middle Temporal Gyrus; ITG, Inferior Temporal Gyrus; SFG, Superior Frontal Gyrus; MFG, Middle Frontal Gyrus; IFG, Inferior Frontal Gyrus; SOG, Superior Occipital Gyrus; MOG, Middle Occipital Gyrus; IOG, Inferior Occipital Gyrus; CS, Calcarine Sulcus; Fus, Fusiform gyrus; Cing, Cingulate gyrus; preC, precentral gyrus; postC, postcentral gyrus; ↓, decrease (of volume when not specified); ↑, increase (of volume when not specified); R, Right; L, Left; CI, Cochlear Implant; Rw as, Rightward asymetry; Lw as, Leftward asymetry; CMV, Cytomegalovirus